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ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B. Tech (Civil Engineering)
Name of the Paper: Fluid Mechanics
Time: 3 Hours

Semester: III
Paper Code: TCE 301
Maximum Marks: 100

Note:

All questions are compulsory

Answer any two sub questions among a, b & c in each main questions

Total marks in each question are twenty

Each question carries 10 marks

Q1

(20 marks)

- (a) Explain viscosity. Compare kinematic and dynamic viscosity and discuss methods to measure viscosity.
- (b) Derive the relation for excess pressure inside a soap bubble of radius r considering surface tension. **CO1**
- (c) Compute capillary rise of water in a glass tube of diameter 3 mm. Take surface tension $\sigma = 0.075 \text{ N/m}$ and contact angle $\theta = 0^\circ$; $\rho_{\text{water}} = 1000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$.

Q2

(20 marks)

- (a) Distinguish between steady/unsteady, uniform/non-uniform, Rotational/Irrrotational and laminar/Turbulent flows.
- (b) Derive Bernoulli's equation from Euler's equation for inviscid flow along a streamline stating assumptions. **CO2**
- (c) Given velocity components $u = 2x$, $v = -2y$ (units m/s), check continuity and compute acceleration at point (1,1).

Q3

(20 marks)

- (a) Explain the working principle of a Venturimeter. Discuss assumptions and sources of error.
- (b) Derive equation for discharge over a broad-crested weir (state assumptions). **CO3**
- (c) Find discharge through a 60° V-notch if head over crest $h = 0.25 \text{ m}$. Use $C_d = 0.62$ and $g = 9.81 \text{ m/s}^2$.

Q4

(20 marks)

- (a) Explain laminar and turbulent flow regimes and discuss physical mechanisms that trigger transition. Mention Reynolds experiment and its practical importance.
- (b) Derive velocity profile for laminar flow in circular pipe and show mean-to-max velocity relation. **CO4**
- (c) A steel ball of radius 1 cm falls through oil of density 900 kg/m^3 and dynamic viscosity $\mu = 1.5 \text{ Pa}\cdot\text{s}$. Using Stokes' law, estimate terminal velocity. Take $g = 9.81 \text{ m/s}^2$.

Q5

(20 marks)

- (a) Discuss major and minor energy losses in pipe networks. Explain how hydraulic gradient line (HGL) and energy grade line (EGL) change along a pipeline with distributed friction and a sudden contraction.
- (b) Derive pressure rise expression for sudden valve closure in elastic pipe.
- (c) Calculate maximum pressure rise if wave speed $c = 1000 \text{ m/s}$ and $V = 1.8 \text{ m/s}$ (use $\rho = 1000 \text{ kg/m}^3$).

CO5